

PRE-READ · ADVANCED ANALYTICS

The Zillow Story

How Zillow Put a Price on Every American Home, and Then Paid for It

A Seattle start-up used data to break the hold of property brokers over America's housing market. Fifteen years later, the same data pulled it into a loss of more than US\$400 million in a single quarter.

Note: This pre-read document is based on several case studies and publicly available documents (See References section).

It is February 2006. Across the United States, curious homeowners are typing their own address into a new website. Within seconds a number appears on the screen: what their house is worth. No broker, no appraiser, no fee. So many people try it on the first morning that the site crashes.

The website was Zillow. Over the next decade it became the country's most visited property portal, the place Americans went first to look at homes and check prices. Then, in November 2021, it shut down an entire business, put about 7,000 houses up for sale and let go of a quarter of its employees.

For anyone studying analytics, Zillow offers two lessons in one story. The first half shows how data beat the gut feel of an entire industry. The second half shows what happens when a model built for one job is handed a very different one.

From Expedia to Zillow

Zillow's founders had done this once before. Rich Barton and Lloyd Frink were Microsoft executives who had started Expedia, the online travel site that let ordinary people book flights and hotels without going through a travel agent. Barton's belief was simple: if the internet could open up travel, it could open up housing too.

The frustration was personal. Recalling 2005, Barton later said that the web had been around for almost ten years, yet he still could not find photographs, full listings, prices or even the address of a house he wanted to buy. The question he wanted answered was the one every buyer asks: what is this house worth, and how much should I offer?

Zillow was registered in Washington state in December 2004. Barton and Frink put in their own money and ran the company from a small office in downtown Seattle with about 20 people. The early team also included Spencer Rascoff, who had co-founded the travel site Hotwire.com.

The first idea did not work. For almost a year the team tried to sell homes through online auctions and got nowhere. The turning point, as the founders tell it, came over a weekend exchange of emails. Google Maps had just started showing satellite pictures of every neighbourhood in America. One colleague wondered whether they could put a price on every rooftop in those pictures. That question became the Zestimate.

The site went live on 8 February 2006 with a free home-value tool that anyone could use, along with past sale records and charts comparing price rises across towns. It collapsed on the first morning under the traffic. By one account, more than a million people visited in the first three days. A San Francisco newspaper jokingly described it as an addiction for people obsessed with property.

ZILLOW'S JOURNEY AT A GLANCE

Year	What happened
2004	Company registered; founders fund it themselves
2006	Website launches with the Zestimate; raises US\$25 million
2008	Launches a home-loan marketplace and an iPhone app
2010	Makes a profit for the first time; Spencer Rascoff becomes CEO
2011	Lists on the Nasdaq stock exchange
2012–2013	Buys rental and city-focused sites HotPads and StreetEasy
2015	Buys its biggest rival, Trulia, for about US\$2.5 billion
2018	Starts buying and selling homes itself through Zillow Offers
2019	Barton returns as CEO
2021	Shuts Zillow Offers and cuts about a quarter of its staff

When brokers held all the cards

Anyone who has hunted for a flat in an Indian city through a broker will recognise the America that Zillow walked into. The buyer saw what the broker chose to show. The price was whatever the broker said was fair. There was no easy way to check.

In the US, homes for sale were recorded in local databases called Multiple Listing Services, or MLS. There were about 500 of them, most run by associations of property brokers, known there as realtors. To get in, a broker usually had to join the National Association of Realtors and a local association. Brokers treated these listings as a private asset. For their access and advice, they typically charged a commission of around 6% of the sale price.

Pricing ran on experience and instinct. A seller found out what a house was worth from a broker's comparison with nearby sales, or by paying for a formal valuation. A buyer had no independent way to test whether an asking price was reasonable. That gap in information was where the broker's power came from.

There was already a big online player. Realtor.com, the official website of the realtors' association, had been running since the late 1990s. But it showed only homes that were on sale, carried little wider market information and was not updated often. Its operator was also caught up in an accounting scandal in the early 2000s. Industry watchers later concluded that Realtor.com wasted its head start.

	The old way	Realtor.com	Zillow
Homes you could see	Whatever your broker searched	Homes on sale	Over 100 million homes, on sale or not
Price information	Broker's opinion or a paid valuation	Asking prices only	A free estimate for almost every home
Who held the data	Brokers	The realtors' association	The public
How decisions were made	Experience and local instinct	Listings plus broker advice	Data, past sales and a model

Zillow turned this upside down. It let buyers look at more than 100 million homes, whether or not they were for sale. Just as cleverly, it presented itself as a way for brokers to advertise, not as a threat to replace them. That kept the people who controlled the listings on its side.

The Zestimate: a price tag on every rooftop

Imagine being able to open an app, point it at any flat in Pune or Bengaluru and see what it is worth, whether or not it is on sale. That is what the Zestimate did for Americans in 2006. It took a judgement that had belonged to brokers and made it a public number.

The Zestimate is what analysts call an automated valuation model. It works out a home's likely value from public records, property tax assessments, the prices at which similar homes nearby have sold, and details such as floor area and location. Zillow has always said it is a starting point, not a formal valuation.

At launch it covered about 40 million homes and was refreshed once a month. It now covers more than 100 million homes and updates several times a week. In 2011 Zillow added a rent estimate for about 90 million homes.

It was far from perfect at first. Stan Humphries, Zillow's chief analytics officer, has said the early Zestimate was often well off the mark, with a typical error of about 14%. A 2007 review by the Wall Street Journal found the estimates were frequently within a few per cent of the actual price, but badly wrong when they missed. A housing advocacy group even complained to the US consumer protection regulator in 2006. The regulator decided not to investigate.

The errors did not stop people from using it. One analysis by Harvard Business School students argues that the Zestimate was designed to get people talking rather than to be precise, and that its real pull was that it covered almost every house in the country. Keep that point in mind. The Zestimate's job was to bring visitors to the website. It was never meant to decide how much money to pay for a house.

Zillow's data team kept improving the model with every new sale. By 2017 the Zestimate was built on 7.5 million statistical and machine learning models, each looking at hundreds of details about a property. That year

Zillow went further and threw the problem open to the world. The Zillow Prize, run on the data science platform Kaggle, offered US\$1 million to anyone who could beat Zillow's own model.

More than 3,800 teams from 91 countries took part over nearly two years. The winners, a three-member team from Morocco, the US and Canada who had never met in person, beat Zillow's model by about 13%. The prize was awarded in January 2019.

Zillow then rebuilt the Zestimate from the ground up. In June 2021 it replaced thousands of separate regional models with a single national deep learning model, which it called the Neural Zestimate. The new model understands location at several levels at once, from the street to the state, accounts for seasonal swings in prices, and gives a price range rather than a single figure. Within a year, Zillow said, errors had fallen by more than 15% nationally.

Period	Typical error reported	What was measured
2006 launch	About 14%	All homes estimated
2017	About 5%	As reported by Zillow when the prize began
January 2019	4.5%, aiming for under 4%	Nationwide error rate
2023	7.49%	Homes not on sale only

These figures are not strictly comparable, because they measure different groups of homes. A home that is listed for sale is easier to value, since the asking price itself is a clue. Zillow now says its estimate is usually within about 2% for homes on sale and about 7% for those that are not.

Giving data away, and still making money

Zillow's business model sounds odd at first. It gave its most valuable information to the public for free. The money came from the other side of the market: brokers who wanted to reach all those visitors.

It worked like a cycle. Free estimates and listings pulled in the biggest audience in American real estate. Brokers put their listings and advertisements where the buyers were. Their payments funded more data and better models, which pulled in still more visitors. Brokers who stayed away risked losing clients to those who joined, so the pressure to sign up kept growing.

The foundation was data that nobody else had put together in one place. Zillow built a database of about 160 million American homes from listing services, local government property records, tax data and home-loan information. Unlike the listing services, it included homes that were not on sale, their full sale history and price forecasts. When its main bulk data supplier refused to renew a contract in 2016, Zillow went out and signed deals directly with individual listing services and local record offices to protect that base.

Homeowners and listing brokers pay nothing to appear on Zillow. The revenue comes from a programme called Premier Agent. Brokers pay to be shown alongside listings in the postal areas they choose, and when a buyer

clicks "Contact agent", the enquiry goes to them as a sales lead. In 2016 alone Zillow said it passed on 16.9 million such leads. Premier Agent and related services still bring in about 70% of the company's revenue.

Zillow also used its own data to fix its own marketplace. After merging with Trulia in 2015, it found that buyers who contacted brokers heard back only about half the time. It changed how its sales team was rewarded, moving away from signing up as many brokers as possible and towards brokers who earned more and served buyers better.

Acquisitions widened its reach. Zillow bought rental and city-specific websites, and in 2015 swallowed its largest rival, Trulia, for about US\$2.5 billion. Together the two accounted for roughly 70% of online property searches in the US.

ZILLOW BY THE NUMBERS

Measure	Zillow	For comparison
Monthly users, 2016	About 115 million across its sites	Realtor.com: about 37 million
Average monthly visitors, 2018	157 million	Ranked 24th among all US websites
Revenue, 2018	US\$1.3 billion, mostly from broker advertising	—
Share of US property website visitors, February 2025	66% (company's own figure)	—

By 2018 Zillow had become the default place Americans went to browse homes and check prices. That position rested on a data asset and a model that were good enough to keep people coming back. It was exactly this success that tempted Zillow into its next move.

The big bet: buying houses itself

In April 2018 Zillow stopped being only a website. Through a new business called Zillow Offers, it began buying homes directly from owners, repairing them and selling them on. The industry calls this "iBuying", short for instant buying.

On paper it made sense. Many owners wanted to sell but disliked the slow, uncertain traditional process. Zillow Offers promised a cash offer within 48 hours for homes that met its criteria. Zillow had the biggest property database in the country, a strong data science team and the most trusted price estimate in America. Why not use all that to buy houses? Several rivals, led by a start-up called Opendoor, were already doing it.

Zillow began carefully. It tested the idea in two cities in 2017 and launched properly in Phoenix, Arizona, in 2018. Pricing combined the machine and the human. The algorithm suggested a value, Zillow's pricing experts

adjusted it for local conditions, and its renovation teams subtracted the expected cost of repairs. In that first year it bought 686 homes, sold 177, earned US\$52 million in revenue and lost US\$27.2 million.

The mood changed when Rich Barton returned as chief executive in February 2019. He set Zillow Offers a target of US\$20 billion in annual revenue within three to five years. Barton believed iBuying was the future of real estate, and he saw Opendoor, which had raised US\$1.3 billion from investors by mid-2019, as a threat to Zillow's very survival. Meeting the target meant buying and selling about 15,000 homes every quarter.

Period	Homes bought	Homes sold
2018 (full year)	686	177
2019 (full year)	6,512	4,313
2020 (full year, including a Covid pause)	4,162	5,337
April–June 2021	3,805	2,086
July–September 2021	9,680	3,032

Zillow expanded from 5 to 23 cities in 2019, but it remained far short of the pace it needed. Then Opendoor listed on the stock market in late 2020 at a valuation of nearly US\$18 billion, and the pressure to catch up grew.

Where it all went wrong

In the July–September quarter of 2021, Zillow Offers lost US\$421 million. On 2 November 2021 the company announced it was closing the business and laying off about 25% of its staff. It was still sitting on roughly 7,000 houses. Barton put the blame on the algorithm, saying that home prices had proved far harder to predict than Zillow had expected.

Academics who later studied the collapse say that is only half the story.

Project Ketchup

By early 2021 Zillow Offers was well behind its buying targets. To speed things up, the company launched an internal drive named, fittingly, Project Ketchup. Four changes made the difference.

First, for some homes, Zillow simply used the Zestimate as its cash offer. Second, its pricing experts were no longer allowed to adjust the algorithm's figures and were asked to stop questioning them. Third, to win more deals, Zillow sometimes raised its offers by thousands of dollars above what the algorithm itself had suggested, a practice staff called "offer calibration". Fourth, it cut its fee to sellers to 1%, against Opendoor's 5%.

It worked, at least on volume. Between July and September 2021 Zillow bought 9,680 homes, more than in the previous five quarters put together. But it sold only a fraction of them, and the unsold houses piled up on its books.

The wrong question for the model

The deeper problem was that buying a house asks a different question from the one the Zestimate was built to answer.

	Zestimate on the website	Price for a cash offer
Question	What is this home worth today?	What will it sell for months from now, after repairs?
How accuracy was checked	Against the price on the day of sale	Needs a forecast for a sale date nobody knows yet
Cost of a mistake	A visitor sees a wrong number	Zillow loses its own money
Other things that matter	Very little	Loan rates, repair costs, contractor availability, time to sell

A website estimate only has to be roughly right today. A purchase price has to anticipate what the market will do between the day of buying and an unknown day of selling, and the longer that gap, the bigger the uncertainty. Zillow was also new to this business. Its rivals had spent years learning how to price homes for resale.

A spiral on the ground

The buying spree ran straight into a shortage of workers in 2021. Zillow's network of contractors could not repair homes fast enough. When the company realised it had been overpaying, it cut renovation budgets. Many contractors, who also worked for rivals that paid better and were easier to deal with, pushed Zillow's jobs to the back of the queue or walked away. Houses sat unsold for longer, and the gap between what Zillow paid and what it could ask was far worse than at Opendoor.

What stands out is that Zillow lost money while American house prices were rising. Its competitors faced the same market, but they had more experience and grew more slowly. None suffered losses on the same scale, and some are still in the business today.

Zillow lost money while American house prices were rising.

Why Zillow Offers failed

The researchers conclude that Zillow's failure came from a mix of weak price forecasting and poor strategic and operational choices. It chased rapid growth in a business it did not understand, and it removed the human experts who had acted as a check on a model that was still learning. Investors punished the company heavily that week, and shareholders later filed a lawsuit claiming its executives had hidden the problems before the shutdown.

Discussion questions

Use the hints if a question stalls. They point to the part of the story that carries the answer, not to the answer itself.

1

What made Zillow succeed in 2006? What problem did it solve?

- Think about who held the information before 2006, and what an ordinary buyer could actually check.
- The Zestimate was often 14% off. Why did that still matter to households?

2

How did Zillow use information (data) to help customers make better decisions?

- Consider coverage: more than 100 million homes, including those not for sale.
- A free estimate gave a seller a starting point instead of one broker's opinion.

3

Do you agree that in most cases data leads you to make better decisions than gut feel or experience?

- Where does the same decision repeat often enough for a model to learn from the result?
- Name one decision in this story where judgement was clearly the better guide.

4

Does more data availability generally tend to lead to better decisions?

- Zillow had the largest property database in America and still lost US\$421 million.
- Compare the two columns in the table on the Zestimate and the cash offer.

5

In your opinion, what was the biggest mistake in Project Ketchup?

- Four changes were made: the Zestimate became the offer, experts could no longer adjust it, offers were raised above the model's figure, and the seller fee was cut.
- Which of the four would you undo first, and what would it have saved?

References

This story is compiled from published case studies, company disclosures and press reports. The account of the Zillow Offers collapse, including Project Ketchup and the purchase and sale figures, draws mainly on the first reference below.

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A NOTE ON FIGURES

The 2024 teaching case gives two different sales figures for July–September 2021 (2,086 and 3,032) and two different falls in Zillow's market value (over 37%, and 80% in one week). This story uses 3,032 homes sold and avoids quoting the market-value fall. Check Zillow's third-quarter 2021 shareholder letter before using these numbers in an assessment.